Erfan Pirjahandideh

Applied AI & Computer Vision

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PROFILE

Applied AI and automation practitioner with current computer vision research experience at Fraunhofer IGD and ongoing M.Sc. studies in Artificial Intelligence and Machine Learning at TU Darmstadt (expected 2027). Previously spent eight years in telecommunications IT, building operational tools, improving service workflows and leading technical teams. Independent projects demonstrate held-out ML evaluation, simple baselines, human-reviewed automation and reproducible experiments.

Looking for applied AI, AI automation and ML engineering roles, or a master's thesis in brain research or neurotechnology. Available now for full-time roles. Based in Darmstadt; open to Berlin, Frankfurt and remote roles based in Germany.

RESEARCH & PROFESSIONAL EXPERIENCE

Computer Vision & AI Research Assistant · Fraunhofer IGD, Darmstadt, Germany — Feb 2026 – present

- Contribute to image-synthesis and biometric-evaluation experiments in an ongoing research project, with an emphasis on reproducible pipelines, controlled comparisons and documented evaluation.
- Work with Python, PyTorch and GPU computing through Slurm; document evaluation methods and their limits.

QA Team & Project Lead | Senior Developer Expert · NAK (MCCI project, telecommunications) — 2022 – 2024

- Led the implementation of a monitoring system covering more than 30 services.
- Developed a Chrome ticketing extension (jQuery, HTML, CSS) that reduced ticket-resolution time by 25%.

Service Desk & Service Management Team Leader · Behsa (MCCI project, telecommunications) — 2018 – 2022

- Set up and led a seven-person service desk handling more than 2,000 tickets a month; integrated the MCCI ticketing system with Jira.
- Introduced ITIL-aligned workflows that improved response times by 40%; coordinated incident and request resolution against SLAs.

First Technical Support & Helpdesk Engineer · AHS (MTN project, telecommunications) — 2016 – 2018

- Automated daily management reports used by senior management and the CIO, improving report accuracy by 30%.
- Built Chrome extensions and service-desk dashboards that reduced manual workload by 20%; built Telegram bots for automated report updates.

Freelance Software Developer & Web Designer · Independent — 2005 – present

- Build web applications, responsive websites, dashboards and workflow automations with PHP, WordPress, Joomla, JavaScript and Python, using AI-assisted development (Cursor, ChatGPT/Codex).

SELECTED INDEPENDENT PROJECTS

Brain Signals Lab — EEG motor imagery · *Completed evaluation · Public dataset* · github.com/erfanpj/brain-signals-lab

- Tested EEG classification of imagined left- vs right-fist movement on public PhysioNet data: 50 participants, 2,245 usable trials, five participant-held-out folds; band and settings chosen on training participants only.
- CSP + LDA reached 61.8% mean accuracy versus a 50.4% majority-class baseline (participant median 57.8%); a reduced EEGNet variant reached 53.9%. Results apply to this evaluated dataset subset, not to EEG decoding in general.

OpsDesk AI — human-reviewed support automation · *Local prototype · Browser simulation* · github.com/erfanpj/opsdesk-ai

- Built a local support-assistance workflow: handbook retrieval (keyword, BM25, dense, hybrid), proposals from a small local language model, human approval and mock actions.
- Editing a proposal invalidates its approval. On the supported local service path, repeated execution with the same approval ID does not repeat the mock action. Review and execution events are recorded in a hash-chained audit log. The FastAPI review UI and MCP adapter share the same service.
- On 60 held-out fictional tickets, rules produced 37 fully correct proposals versus 25 from the local model, so I kept the rules as the default.

Breast Cancer Classifier Comparison · *Completed evaluation · Not a clinical tool* · github.com/erfanpj/Breast_Cancer_Project

- Leakage-aware rework with nested cross-validation; on 143 held-out samples a tuned MLP (F1 0.950) matched an untuned logistic-regression baseline (0.951).

Exploratory Alzheimer's Data Analysis — Julia · *Educational analysis · Synthetic data* · github.com/erfanpj/exploratory_alzheimers

- Reproducible descriptive-analysis workflow with validation, summary tables and 49 automated tests on 2,149 synthetic records. The data does not represent real patients; the results are not medical findings.

Reachboard — Django Workflow Showcase · *Fictional-data application · Browser simulation available* · github.com/erfanpj/reachboard-showcase

- Django workflow demonstration with server-side pricing, ownership checks, controlled state changes and an audit history; repeated submissions do not create duplicate history entries.

I used AI coding assistants on these independent projects; each README states what is fictional, local-only or simulated.

EDUCATION

M.Sc. Artificial Intelligence and Machine Learning · TU Darmstadt, Germany — 2024 – present

In progress; expected graduation 2027.

B.Sc. Software Engineering · Islamic Azad University — 2009 – 2013

Associate Degree in Software Engineering · Islamic Azad University — 2006 – 2009

TECHNICAL SKILLS

Applied AI & automation: LLM-assisted workflows, retrieval (keyword, BM25, dense, hybrid), local language models, MCP, human-in-the-loop approval, audit logging, FastAPI; AI-assisted development with Cursor and ChatGPT/Codex.

ML & computer vision: Python, PyTorch, CUDA, OpenCV, scikit-learn, Hugging Face Diffusers, biometric evaluation (EER, FMR/FNMR).

Evaluation & research engineering: Held-out and nested cross-validation, simple baselines, reproducible pipelines, dataset-integrity checks, experiment documentation.

Software & infrastructure: Python, Julia, SQL, JavaScript, PHP, Groovy; Django, Celery, Laravel, PostgreSQL, MySQL, APIs; Git, Linux, Docker, Slurm, Playwright, Power BI.

Process & delivery: ITIL, Jira, SCSM, SLA/OLA, QA and software testing, workflow automation, stakeholder coordination, team leadership.

PUBLICATION & AWARD

Co-author, “Personality Recognition from Text Based on Deep Learning with Recurrent Neural Network” (conference paper, 2017).

CEO Special Award for Innovation, MTN, April 2018.

LANGUAGES

Persian/Farsi — Native | English — Advanced | German — A1